

Regional Resource Management for Service Provisioning in LEO Satellite Networks: A Topology Feature-Based DRL Approach

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Abstract—Satellite networks with wide coverage are considered natural extensions to terrestrial networks for their long-distance end-to-end (E2E) service provisioning. However, the inherent topology dynamics of low earth orbit satellite networks and the uncertain network scales bring an inevitable requirement that resource chains for E2E service provisioning must be efficiently re-planned. Therefore, achieving highly adaptive resource management is of great significance in practical deployment applications. This paper first designs a regional resource management (RRM) mode and further formulates the RRM problem that can provide a unified decision space independent of the network scale. Subsequently, leveraging the RRM mode and deep reinforcement learning framework, we develop a topology feature-based dynamic and adaptive resource management algorithm to combat the varying network scales. The proposed algorithm successfully takes into account the fixed output dimension of the neural network and the changing resource chains for E2E service provisioning. The matched design of the service orientation information and phased reward function effectively improves the service performance of the algorithm under the RRM mode. The numerical results demonstrate that the proposed algorithm with the best convergence performance and fastest convergence rate significantly improves service performance for varying network scales, with gains over compared algorithms of more than 2.7%, 11.9%, and 10.2%, respectively.

Index Terms—Satellite networks, resource management, service performance optimization, topology feature learning.

I. INTRODUCTION

Nowadays, the rapid development of the economy and society has brought about earth-shaking changes in the lifestyles of people all over the world. Various emerging applications, such as augmented reality, virtual reality, and Internet of Things (IoT), are needed by all kinds of users in all regions [1]. However, mobile communication systems that still rely on terrestrial infrastructure have never been able to achieve global ubiquitous connectivity and seamless services [2]. To this end, researchers of the sixth-generation (6G) mobile communication system are working hard to achieve this grand goal and have driven the deployment of satellite networks based on

low earth orbit satellites (LEOs), such as OneWeb, Starlink, and Telesat [3]. Satellite networks with wide coverage are considered irreplaceable in the long-distance transmission of services compared with existing terrestrial networks or even air networks. Therefore, it is very important and highly favored by users for satellite networks for end-to-end (E2E) service provisioning.

In this case, users will access the LEO through a direct connection mode that is different from the traditional satellite network that provides information exchange between ground stations, and E2E service provisioning will be achieved using the satellite network as the service-bearing entity between any two nodes, such as user-to-user, user-to-server, etc [4]. However, due to the inherent orbital deployment and high-speed movement of LEOs, inter-satellite links (ISLs) are intermittently connected, which means that the network topology is highly dynamic to inevitably re-plan the resource chains for E2E service provisioning. Furthermore, the scale of satellite networks is highly differentiated, ranging from tens to tens of thousands of LEOs, which means that resource management algorithms should be highly adaptable in practical deployment applications.

Resource management for E2E service provisioning in satellite networks has attracted more attention recently [5]–[8]. The resource evolution relationship based on the satellite network topology was modeled as a time-expanded graph, and the service flow optimization problem on this graph was solved by leveraging the proposed iterative heuristic algorithms in [5], which is a service provisioning scheme that directly plans E2E transmission resource chains. To combat the dynamic network environment, [6] applied deep reinforcement learning (DRL) to service provisioning to optimize resource management strategy by selecting candidate resource chains to meet resource constraints. However, the path-level decision models in [5] and [6] are difficult to cope with the varying network scale, and in light of the fixed characteristics of the neural network output dimension, the candidate resource chains are calculated in advance and the number cannot be changed, which hinders the practical application value of the algorithms. In [7], the E2E service provisioning was formulated as a series of next-hop selection

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processes, and a resource chain planning algorithm based on network topology feature extraction was proposed. Similarly, [8] proposed a DRL-based E2E service provisioning algorithm that can simultaneously make next-hop decisions for multiple services. However, the input of a global network structure in [7] and the actions of fixed dimension related to the number of service requests in [8] are bound to lead to retraining under varying network environments.

This paper investigates the resource management for E2E service provisioning in satellite networks to alleviate the limitations of network environment uncertainty on the practical deployment and application of the algorithm. There are two challenges as follows:

- 1) How to formulate a resource management problem to ensure the unification of decision spaces under varying network scales?
- 2) How to design a resource management algorithm to alleviate the impact of dynamic network environments in its adaptability?

To solve the above challenges, we propose a topology feature-based dynamic and adaptive resource management (TF-DARM) algorithm to ensure E2E service provisioning performance in different satellite networks. The contribution of our work can be summarized as follows:

- 1) **Regional resource management mode:** We start from the orbital deployment and movement of LEOs and clarify the similar topology features of different satellite network environments. Based on this feature, we design a regional resource management (RRM) mode that is different from conventional multi-resource chain E2E service provisioning and further formulate the RRM problem that can provide a unified decision space in different network scales.
- 2) **TF-DARM algorithm:** To combat the dynamic network environments and avoid retraining, we develop the TF-DARM algorithm to obtain a trained model independent of the network scale. Specifically, the algorithm adopts a generalized action space to take into account the fixed output dimensions of the neural network and the changing resource chains for E2E service provisioning in various network environments. Furthermore, the state with service orientation information and the phased reward function are designed to progressively guide service requests to approach the destination node.

The remainder of this paper is organized as follows. Section II presents the system model. The RRM problem is formulated in Section III. In Section IV, we convert the problem into a Markov Decision Process (MDP) and propose the TF-DARM algorithm to solve it. The numerical results are shown in Section V, and finally, we conclude the paper in Section VI.

II. SYSTEM MODEL

In this section, we first describe the network model, and then introduce the service request model, and delay model, respectively.

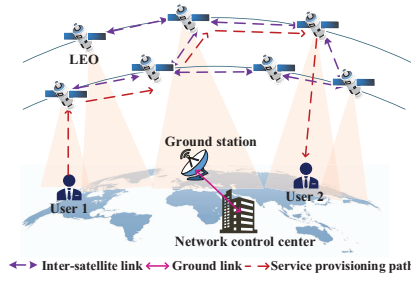


Figure 1. Illustration of a satellite network scenario and service request transmitting.

A. Network Model

We consider a typical satellite network scenario for E2E service provisioning, which mainly includes a set of LEOs $\mathcal{L} = \{i | i = 1, 2, \dots, |\mathcal{L}|\}$ and one network control center (NCC), as shown in Figure 1. Furthermore, this paper considers the satellite network realizes service bearing between the source and the destination nodes of service requests. For example, as shown in Figure 1, “User 1” transmits its services to “User 2” through the satellite network. It should be noted that the ground stations summarize resources and service request information from LEOs to the NCC for training resource management models, and we assume that the effectiveness of network resources and service requirement information can be guaranteed to study the impact of resource management strategies on the service performance of satellite networks.

Due to the orbiting movements, the connection relationship between two LEOs is time-varying. We can observe that the topology of satellite networks directly affects resource chains for E2E service provisioning. Meanwhile, we also observe that satellite networks are different from terrestrial and air networks. The characteristics of their orbital deployment make the “one-satellite four-links” mode widely used in the establishment of ISLs [9]. Specifically, in light of the laser ISL has been widely used in constellations, such as the Starlink constellation, this paper considers LEOs with laser communication devices to exchange with each other [10]. According to the ISL establishment rule, each LEO is equipped with four laser terminals, which are used to establish two intra-orbit plane laser ISLs and two inter-orbit plane laser ISLs, respectively [11]. The set of directional ISLs is denoted by $ISL = \{(i, j) | i, j \in \mathcal{L}\}$. Furthermore, we divide the planning cycle into a set of time slots, denoted by T , and the interval of each time slot $t \in T$ is fixed as τ .

B. Service Request Model

This paper focuses on the E2E service provisioning process after service requests arrive at LEOs, considers the batch service provisioning of service requests, and sets its batch service period to a time slot [12]. Furthermore, we assume that a batch of service requests with different service deadlines arrives in each service period, and we determine the service deadline of each service request based on its arrival time and delay requirement¹. In addition, the service process of service

¹This paper considers the causality of on-board storage, i.e., service requests in the current time slot arriving are served in the next time slot.

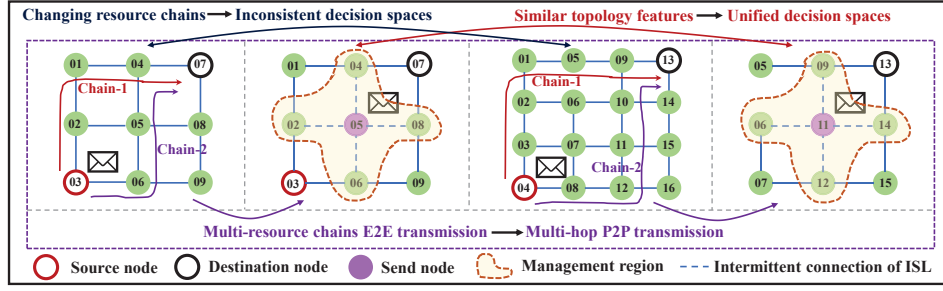


Figure 2. Illustration of resource management mode for E2E service provisioning in satellite networks.

requests may last for multiple time slots, and service requests across time slots will be combined with newly arrived service requests into one batch to be served.

Furthermore, we construct a service request sequence, denoted by $\mathcal{Q} = \{q | q = 1, 2, \dots, |\mathcal{Q}|\}$ and define each service request q denoted as $q = (S_q, D_q, At_q, B_q^{\text{req}}, L_q^{\text{req}})$, where S_q and D_q represent the source and destination node, respectively. At_q is the arrival time. B_q^{req} and L_q^{req} represent the data transmission requirement and delay requirement, respectively.

C. Delay Model

Considering the advantages of satellite networks in long-distance transmission, service requests using satellite network-providing service may go through multiple hops from the source node to the destination node. The delay of the single-hop consists of four parts, expressed as:

$$L_q^i(t) = L_{\text{tran}}^{(i,j)}(q,t) + L_{\text{prop}}^{(i,j)}(q,t) + L_{\text{proc}}^i(q,t) + L_{\text{que}}^i(q,t). \quad (1)$$

Wherein, $L_{\text{tran}}^{(i,j)}(q,t)$ represents the transmission delay, calculated as:

$$L_{\text{tran}}^{(i,j)}(q,t) = \frac{B_q^{\text{req}}}{\mathcal{R}^{(i,j)}(t)}, \quad (2)$$

where $\mathcal{R}^{(i,j)}(t)$ indicates the achievable transmission rate (in bps) of ISL from LEO i to LEO j in the t -th time slot. $L_{\text{prop}}^{(i,j)}(q,t)$ represents the propagation delay, expressed as:

$$L_{\text{prop}}^{(i,j)}(q,t) = \frac{d^{(i,j)}(t)}{C}, \quad (3)$$

where $d^{(i,j)}(t)$ is the propagation distance (in km), and C is the speed of light (in km/s). $L_{\text{proc}}^i(q,t)$ represents the processing delay, and we consider that each service request is compressed when arriving at the LEO to reduce the requirement for the link rate. $L_{\text{que}}^i(q,t)$ represents the processing delay and the queuing delay.

III. FORMULATION

In this section, we first introduce the RRM mode. Then, we present the resource, link, and service provision constraints based on RRM mode. Finally, we formulate the proposed RRM problem based on these constraints.

A. Regional Resource Management Mode

As we all know, trained neural network models can perform stably with excellent performance in similar training environments. However, we also understand that varying network scales have dealt a severe blow to the practicality of DRL-based resource management algorithms. In the resource management problem of satellite networks for E2E service

provisioning, the essence of its inability to cope with uncertainty is the change of candidate resource chains, as shown in Figure 2. To this end, we design the RRM mode to ensure the unification of decision spaces leveraging the local topology feature of satellite networks.

Specifically, as shown in Figure 2, in this mode, conventional multi-resource chains E2E transmission decisions are converted into multi-hop point-to-point (P2P) transmission decisions to complete the service provisioning. We divide the one-hop reachable range centered on the send node into a region and determine the decision space for P2P transmission based on the region. In light of the limited number of transceivers and the widely adopted ISL establishment rules as described in Section II-A, the local topology features of different satellite networks are similar, which ensures that the maximum number of next-hop nodes that can be selected at any send node is consistent. Besides, we also clarify that due to the intermittent connection of ISL, the number of optional next-hop nodes contained in the region changes dynamically. Based on the above analysis, we can conclude that applying the RRM mode to formulate the resource management problem can provide a unified decision space for the solution algorithm based on the DRL framework.

B. RRM-Based Mode Constraints

1) *Communication Resource Constraint*: The total data volume of transmitted service requests cannot exceed the maximum data volume that can be carried by ISL (i,j) , expressed as:

$$\sum_{q \in \mathcal{Q}} B_q^{\text{req}} \cdot \xi_q^{(i,j)}(t) \leq \mathcal{R}^{(i,j)}(t) \cdot \tau, \forall (i,j), t, \quad (4)$$

where $\xi_q^{(i,j)}(t)$ is a binary variable for indicating whether the service request q is transmitted on the ISL (i,j) in the t -th time slot, 1 if q is transmitted on the ISL (i,j) , 0 otherwise.

2) *Link Selection Constraint*: For any one RRM decision-making, this paper does not consider the splitting of service requests, thus, only one ISL can be selected for transmitting service requests, expressed as:

$$\sum_{(i,j) \in \mathcal{ISL}} \xi_q^{(i,j)}(t) \leq 1, \forall (i,j), t. \quad (5)$$

3) *Service Provisioning Constraints*: The E2E service provisioning of any service request that is successfully completed should satisfy: 1) the start node and the end node are the source node and the destination node, respectively, expressed as follows:

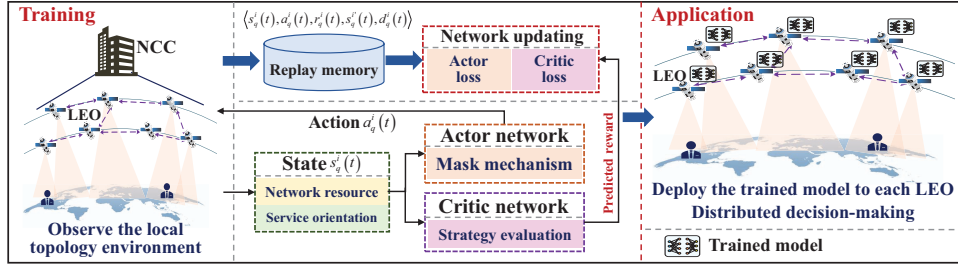


Figure 3. An overview of training and application of the proposed TF-DARM algorithm.

$$\sum_{t \in T(S_{q,j})} \sum_{i \in ISL} \xi_q^{(S_{q,j})}(t) = 1, \forall q, \quad (6)$$

$$\sum_{t \in T(i, D_q)} \sum_{i \in ISL} \xi_q^{(i, D_q)}(t) = 1, \forall q. \quad (7)$$

and 2) the total time from the source to the destination node cannot exceed the delay requirement, expressed as:

$$\sum_{t \in T} \left(\sum_{(i,j) \in ISL} \left(L_{\text{tran}}^{(i,j)}(q, t) + L_{\text{prop}}^{(i,j)}(q, t) \right) \cdot \xi_q^{(i,j)}(t) + \sum_{i \in \mathcal{L}} \left(L_{\text{proc}}^i(q, t) + L_{\text{que}}^i(q, t) \right) \right) \leq L_q^{\text{req}}, \forall q. \quad (8)$$

Otherwise, it will be regarded as a service failure and will be deleted from the on-board storage.

Furthermore, we define a binary variable $\mathcal{N}_q$ to indicate whether the service request q is successfully served, 1 if successfully served, 0 otherwise.

C. Problem Formulation

This work aims to ensure E2E service provisioning performance in satellite networks by optimizing RRM strategies, endeavoring to maximize the number of successfully accomplished service requests while satisfying constraints on the network resources and QoS requirement of services. Mathematically, the RRM problem is expressed as:

$$\begin{aligned} \text{RRM: } \max_{\xi_q^{(i,j)}(t)} \quad & \sum_{q \in \mathcal{Q}} \mathcal{N}_q \\ \text{s.t. } \quad & (4) - (8). \end{aligned} \quad (9)$$

Due to the dynamic network environment, the proposed RRM problem cannot be solved directly by traditional static optimization tools. This paper takes advantage of DRL in combating dynamic network environments and further combines the high adaptability requirements in practical deployment applications to propose the TF-DARM algorithm. The proposed algorithm will be introduced in the next section.

IV. TF-DARM ALGORITHM DESIGN

In this section, the TF-DARM algorithm for E2E service provisioning in satellite networks is designed to solve the formulated RRM problem. Specifically, we first analyze the service provisioning process and model the RRM problem as the MDP. Subsequently, we present the training and application of the TF-DARM algorithm.

A. RRM Problem Conversion

The RRM problem is essentially the process of achieving high-performance E2E service provisioning by allocating the optimal available resources for each service request. During the service provisioning, the P2P transmission decision of a service request is based on the currently available network

resources and the distance between the next-hop node and the destination node and will affect the next network environment, which is an MDP. The main elements of the proposed MDP are shown as follows:

1) *State–Network Resource and Service Orientation*: In the RRM problem, state evolution by time slot is a commonly adopted way [8], and only one-hop transmission of a service request is performed in each time slot, which is not in line with the actual scenario. We design a three-dimensional state evolution way and define the state $s_q^i(t)$ to characterize the available resources of LEO and service orientation, as follows:

$$s_q^i(t) = \left[\bar{\mathcal{R}}^{(i,j)}(t), \eta_q^{(i,j)}(t), \Delta_q^j(t), \chi_q^j(t) \right]_{j \in N_i}, \quad (10)$$

where N_i is the set of next-hop nodes of LEO i , $\bar{\mathcal{R}}^{(i,j)}(t)$ is the normalized available communication resources of LEO i , and $\eta_q^{(i,j)}(t)$ is the probability of successfully transmitting the service request on ISL (i, j) . $\Delta_q^j(t)$ is the service orientation information, i.e., the relative position between the next-hop node j and the destination node D_q , including closing, far away, reaching, etc., expressed as:

$$\Delta_q^j(t) = \begin{cases} 2, & j = D_q, \\ 1, & j \neq D_q \text{ and } j \text{ is close to } D_q, \\ -1, & j \neq D_q \text{ and } j \text{ is not close to } D_q, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

$\chi_q^j(t)$ is the ratio of supply and demand of available communication resources of next-hop nodes, expressed as:

$$\chi_q^j(t) = \begin{cases} 1, & j = D_q, \\ \frac{\sum_{j' \in N_j} \bar{\mathcal{R}}^{(j,j')}(t)}{|\mathcal{Q}_j(t)|}, & j \neq D_q \text{ and } j \in \hat{N}_i(t), \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

where $\mathcal{Q}_j(t)$ is the set of service requests in LEO j in the t -th time slot, $|\cdot|$ indicates getting the number of elements in a set, and $\hat{N}_i(t)$ is the set of optional next-hop nodes.

2) *Action–Regional Resource Management Strategy*: Based on the RRM mode, we design an action space $\mathcal{A}_q^i(t)$ that can take into account the fixed output dimensions of the neural network and the changing resource chains of the dynamic network environments for E2E service provisioning, which corresponds to the all next-hop nodes of LEO i and can be generally expressed as $\mathcal{A}_q^i(t) = N_i \cup \text{None}$, where *None* indicates that no next-hop node is selected, i.e., the service request is not transmitted. Furthermore, due to the intermittent connectivity of ISLs, the set of available actions $\mathcal{A}_q^i(t)$ is

variable and can be determined according to the connection relationship of ISLs, i.e., $A_q^i(t) = \{a_q^i(t) = j \mid j \in \hat{N}_i(t)\} \cup \text{None}$.

3) *Reward-Phased Destination Guidance*: Considering the P2P transmission decision-making, the model of simply giving rewards upon reaching the destination node makes effective guidance information too sparse. To this end, we design the phased reward function matching the service orientation information to progressively guide service requests to approach the destination node, expressed as:

$$r_q^i(t) = \begin{cases} 100 & a_q^i(t) = D_q, \\ \frac{\Delta_q^j(t) \cdot \chi_q^j(t)}{L_q^i(t)} & a_q^i(t) \neq D_q, D_q \notin \hat{N}_i(t), \hat{N}_i(t) \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

To sum up, the RRM problem can be converted into maximizing the long-term cumulative reward, as follows:

$$\max \mathbb{E}_\pi \left[\sum_{t \in T} \sum_{q \in Q} \sum_{i \in P_q(t)} r_q^i(t) \right], \quad (14)$$

where π represents a mapping from $s_q^i(t)$ to $a_q^i(t)$, i.e., $a_q^i(t) = \pi(s_q^i(t))$. $P_q(t)$ indicates the set of LEOs that transmit the service request q in the t -th time slot.

B. Training and Application of the TF-DARM Algorithm

As we mentioned earlier, this paper expects to achieve high adaptability of the algorithm in different network environments and avoid retraining. Therefore, we choose to adopt a centralized training mode with the NCC as the agent to obtain a trained model that is independent of the network scale and introduce the Advantage Actor-Critic (A2C) framework to optimize the parameters of the neural network. The overview of training of the proposed algorithm is shown in Figure 3.

Specifically, the NCC observes the state of the local topology environment centered on any LEO and applies the actor network $\pi_\vartheta(s_q^i(t))$ to select the RRM strategy, where a mask mechanism is designed to ensure the validity of the selected strategy. Then, the NCC collects a series of experience data by interacting with the local topology environments. These experience data consist of the three-dimensional state evolution sequences, and provide rich local topological environment features, which can help the NCC quickly and comprehensively learn the changing network environments. Sample the experience data and apply loss functions to update the parameters of the actor network and critic network, where the critic network, denoted by $V_\varpi(s_q^i(t))$, outputs the predicted reward and is responsible for evaluating the selected strategy. The loss functions are expressed as follows:

$$\mathcal{L}(\vartheta) = -\frac{1}{|\mathcal{M}|} \sum_{s_q^i(t) \in \mathcal{M}} \log(\pi_\vartheta(a_q^i(t) | s_q^i(t))) \cdot \mathcal{W}_q^i(t), \quad (15)$$

$$\mathcal{L}(\varpi) = \frac{1}{2 \cdot |\mathcal{M}|} \sum_{s_q^i(t) \in \mathcal{M}} (R_q^i(t) - V_\varpi(s_q^i(t)))^2, \quad (16)$$

where $\mathcal{M} = \langle s_q^i(t), a_q^i(t), r_q^i(t), s_q^{i'}(t), d_q^i(t) \rangle$ is the minibatch, and $d_q^i(t)$ indicates whether the service request q continues to be served in the t -th time slot. $\mathcal{W}_q^i(t)$ is the

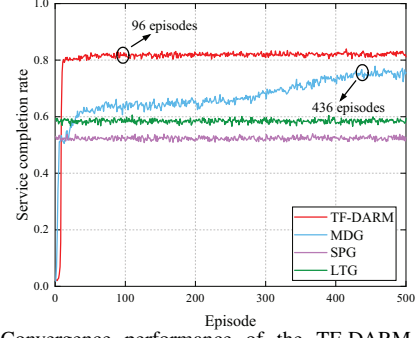


Figure 4. Convergence performance of the TF-DARM algorithm and comparison algorithms.

Temporal-Difference error. $R_q^i(t) = r_q^i(t) + \gamma \cdot V_\varpi(s_q^{i'}(t))$ is the estimated state value, and γ is the discount factor.

In the application phase, as shown in Figure 3, the trained model is deployed to each LEO of the satellite network to make distributed decision-making about RRM strategies.

V. SIMULATIONS

In this section, we give the convergence result and the numerical results for evaluating the performance of the TF-DARM algorithm. For the simulations, the model is trained in a satellite network scenario with 66 LEOs and tested in different satellite networks. Specifically, according to the Iridium communication system, 66 LEOs are distributed over six orbits at a height of 780 km and with an inclination of 86.4° . The configurations of other test networks are set according to the Starlink constellation. The duration of the planning cycle is 1 hour from 30 Dec. 2024 10:00:00 to 30 Dec. 2024 11:00:00. For transmission rate, we set $\mathcal{R}^{(i,j)}(t) \in [5, 10]$ Gbps. Moreover, we set $\gamma = 0.99$, $episode = 500$, $|\mathcal{M}| = 64$, $\tau = 60$ s. The learning rates of actor and critic networks are set $\alpha_\vartheta = 2e^{-4}$ and $\alpha_\varpi = 5e^{-4}$, respectively. Besides, we set that service requests to arrive randomly, with more than 100 arriving for each LEO, $B_q^{\text{req}} = 5$ Gbits and $L_q^{\text{req}} = 5$ s. To compare the performance, three additional approaches are considered:

- **Mismatched Destination Guidance (MDG)**: This approach adopts the framework of the proposed TF-DARM algorithm but does not provide service orientation information that matches the phased reward function of destination guidance, which may lead to long-term exploration and unclear destinations.
- **Shortest Path Greedy (SPG)**: This approach selects the next-hop node closest to the destination node when making a P2P transmission decision for each service request.
- **Least Time Greedy (LTG)**: This approach selects a strategy with the least single-hop delay when making a P2P transmission decision for each service request.

We first evaluate the convergence performance of the TF-DARM algorithm, as shown in Figure 4. It can be seen that at the beginning of the iteration, the TF-DARM and MDG algorithms have a low service completion rate. With the number of iterations increasing, the agent quickly learns

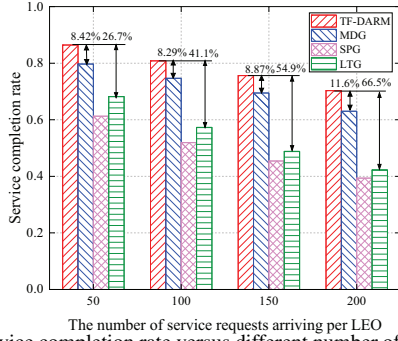


Figure 5. Service completion rate versus different number of service requests.

effective decision-making information and achieves a higher service completion rate than the SPG and LTG algorithms. Besides, the TF-DARM algorithm can provide targeted destination guidance to obtain better convergence performance and a faster convergence rate than the MDG algorithm.

Figure 5 shows the service completion rate under different number of service requests. As we expected, the TF-DARM algorithm achieves the best service performance. Due to a lack of clear service guidance and visionary strategy selection, comparison algorithms have poor service effectiveness and low service completion rates. When the number of service requests arriving per LEO is 50, the service completion rate of the TF-DARM algorithm is higher than 8.42% and 26.7% compared with the MDG algorithm and LTG algorithm, respectively. As the number of service requests increases, the improvement is becoming more and more significant. Besides, since network resources are limited, as the number of service requests increases, the service completion rate of all algorithms gradually decreases.

Finally, we test the service performance of the proposed algorithm under different network scales, as shown in Table I. To ensure the fairness of the test, we set that the number of service requests increases proportionally with the available resources of different network scales. The results indicate that the proposed TF-DARM algorithm exhibits better service performance when generalized to satellite networks with varying numbers of LEOs. Compared with the MDG algorithm, the SPG algorithm, and the LTG algorithm, the TF-DARM algorithm obtains the minimum gains of 2.7%, 11.9%, and 10.2%, respectively. Furthermore, due to the orbital deployment of LEOs, when the number of orbits is small, the ISLs of two LEOs in different orbital planes may rarely be connected, which causes a large number of timeouts for service requests with delay requirements, resulting in poor service performance. With network scale increases, the ISL between the two LEOs has better connectivity. In this case, the service performance of the SPG algorithm has been significantly improved and surpassed the LTG algorithm.

VI. CONCLUSION

In this paper, we explore the topology features of satellite networks and adopt the designed RRM mode to formulate the RRM problem for E2E service provisioning to obtain the

Table I

SERVICE COMPLETION RATE ON DIFFERENT NETWORK SCALES.

Network scale	172	348	720	1584	Minimum gain
Configuration	4*43	6*58	36*20	72*22	
TF-DARM	0.378	0.569	0.837	0.834	/
MDG	0.351	0.541	0.718	0.722	0.027
SPG	0.253	0.450	0.616	0.618	0.119
LTG	0.276	0.461	0.585	0.565	0.102

unified decision space. Subsequently, we model the service provisioning process as the MDP, and based on the A2C framework, we propose the TF-DARM algorithm to combat the dynamic network environments and avoid retraining. The proposed algorithm adopts the three-dimensional state evolution way and leveraging designed generalized action space, it can take into account the fixed output dimension of the neural network and the changing resource chains for E2E service provisioning. Furthermore, the matched design of the service orientation information and phased reward function effectively improves the service performance of the algorithm. Simulation results demonstrate that the TF-DARM algorithm has the best convergence performance and fastest convergence rate and achieves highly adaptive resource management for varying network scales to boost practical deployment applications.

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